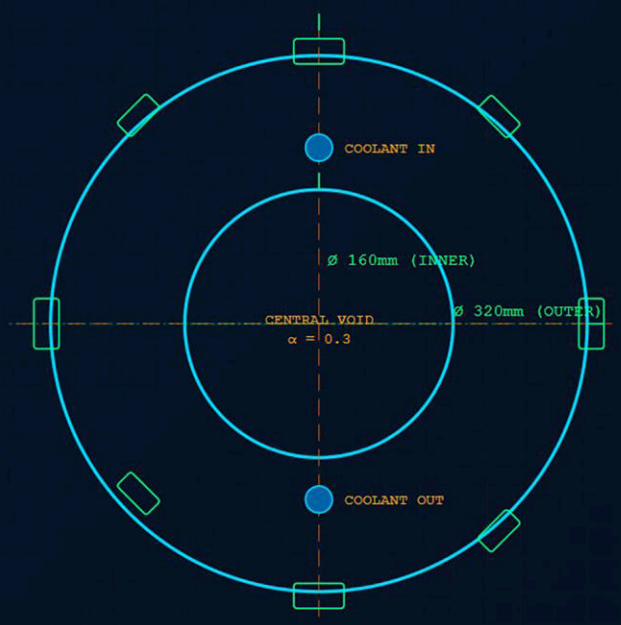
AETHERION MARK 2-V "PROMETEO"

GRADIENT HARVESTING OSCILLATORY SUB-BAND TRANSFER — GENERAL ARRANGEMENT

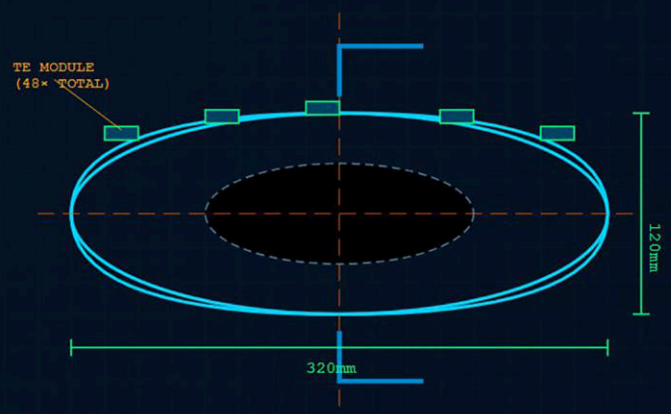
DRAWING: RTM-UFF-AP2V-GA-001
SHEET 1 OF 4
SCALE: AS NOTED

VIEW A: TOP (PLAN VIEW) — SCALE 1:2



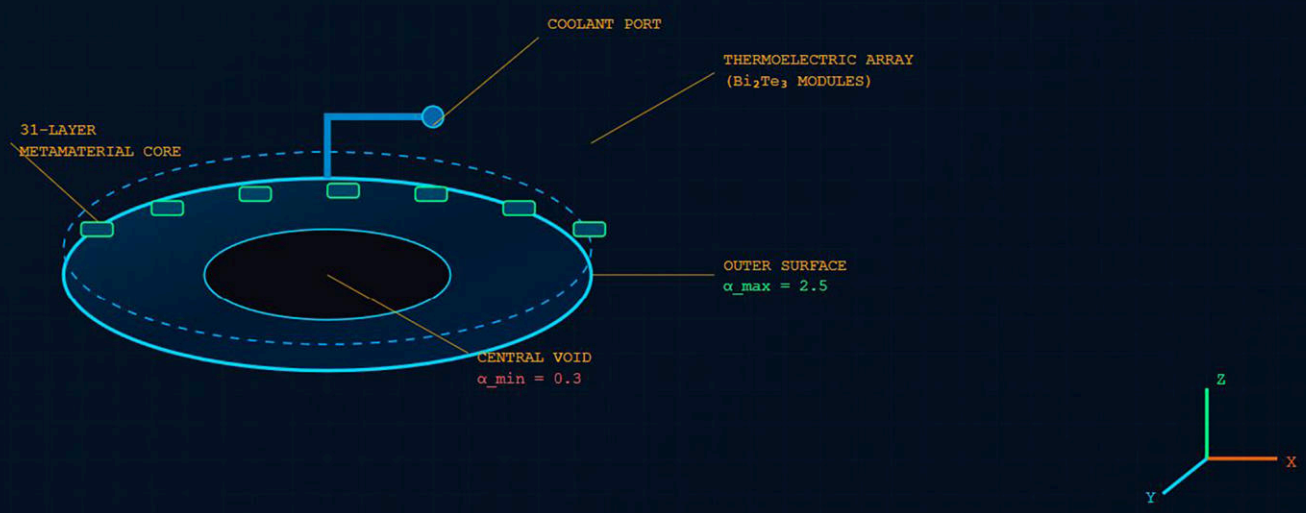
SCALE 1:2

VIEW B: SIDE (ELEVATION) — SCALE 1:2



SCALE 1:2

VIEW C: ISOMETRIC — NOT TO SCALE



NOTES:
1. ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE NOTED
2. TOLERANCES: LINEAR ±0.1mm, ANGULAR ±0.5°
3. SURFACE FINISH: Ra 1.6 µm UNLESS OTHERWISE SPECIFIED
4. MATERIAL: SEE SHEET 3 FOR LAYER SPECIFICATIONS

REVISION HISTORY
REV A — 2026-03-07 — INITIAL RELEASE

DESIGNED: [REDACTED]
CHECKED: [REDACTED]
APPROVED: GHOST PROJECT LEAD

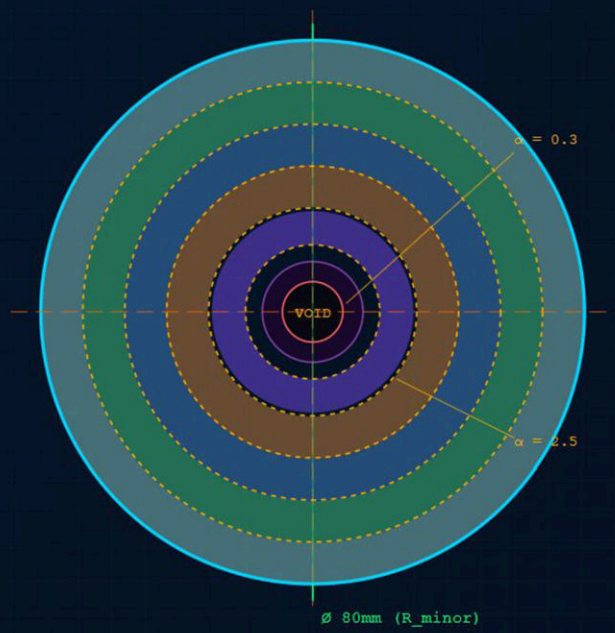
RTM-UFF-AP2V-GA-001
PROMETEO
SHEET 1 OF 4

AETHERION MARK 2-V "PROMETEO"

CROSS-SECTIONAL VIEWS — CORE STRUCTURE

DRAWING: RTM-UFF-AP2V-CS-002
SHEET 2 OF 4
SCALE: AS NOTED

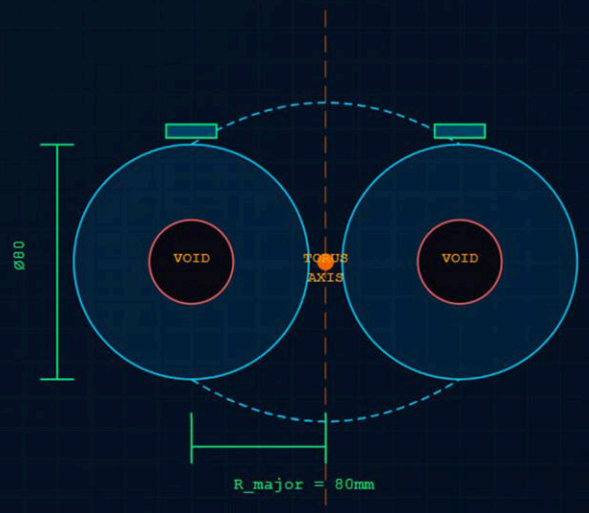
SECTION A-A: POLOIDAL CROSS-SECTION — SCALE 1:1



L1-5: W-Re
L6-10: Nb-Ti
L11-15: Cu-Ni
L16-20: Fe-Co
L21-25: Mn-Cr
L26-31: V-Sc

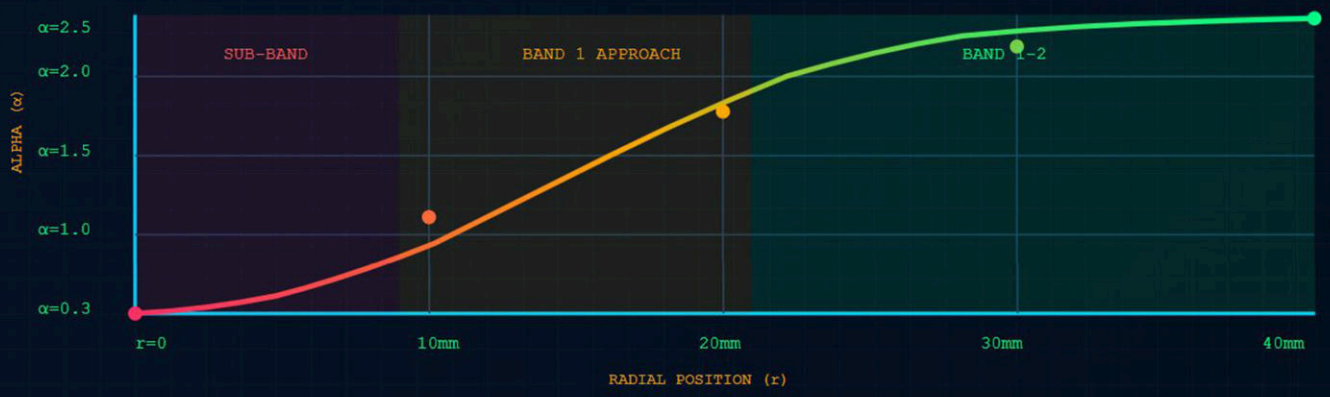
SCALE 1:1

SECTION B-B: TOROIDAL CROSS-SECTION — SCALE 1:3



SCALE 1:3

ALPHA GRADIENT PROFILE — α(r) vs RADIAL POSITION



NOTES:
1. PIEZO INTERFACES SHOWN AS DASHED LINES (30 TOTAL)
2. ALPHA PROFILE: $\alpha(r) = 0.3 + 2.2 \times [1 - (1 - r/R)^{2.5}]^{1.2}$
3. MAXIMUM GRADIENT OCCURS AT $r \approx 15$-25mm
4. GHOST CLAN THRESHOLD ($\alpha > 3.0$) NOT APPROACHED IN STANDARD OPERATION

REVISION HISTORY
REV A — 2026-03-07 — INITIAL RELEASE

DESIGNED: [REDACTED]
CHECKED: [REDACTED]
APPROVED: GHOST PROJECT LEAD

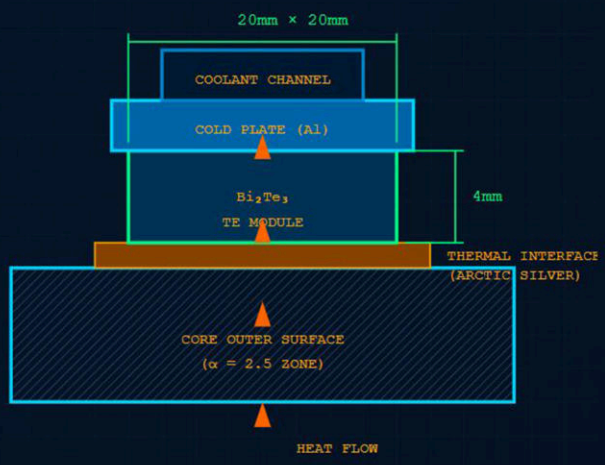
RTM-UFF-AP2V-CS-002
PROMETEO
SHEET 2 OF 4

AETHERION MARK 2-V "PROMETEO"

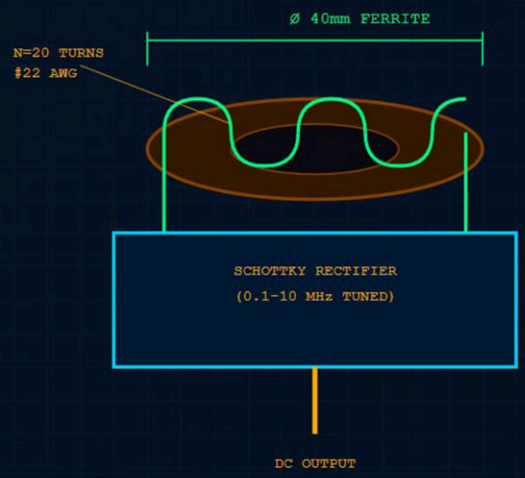
ENERGY HARVESTING SYSTEMS — DETAILED

DRAWING: RTM-UFF-AP2V-EH-003
SHEET 3 OF 4
SCALE: AS NOTED

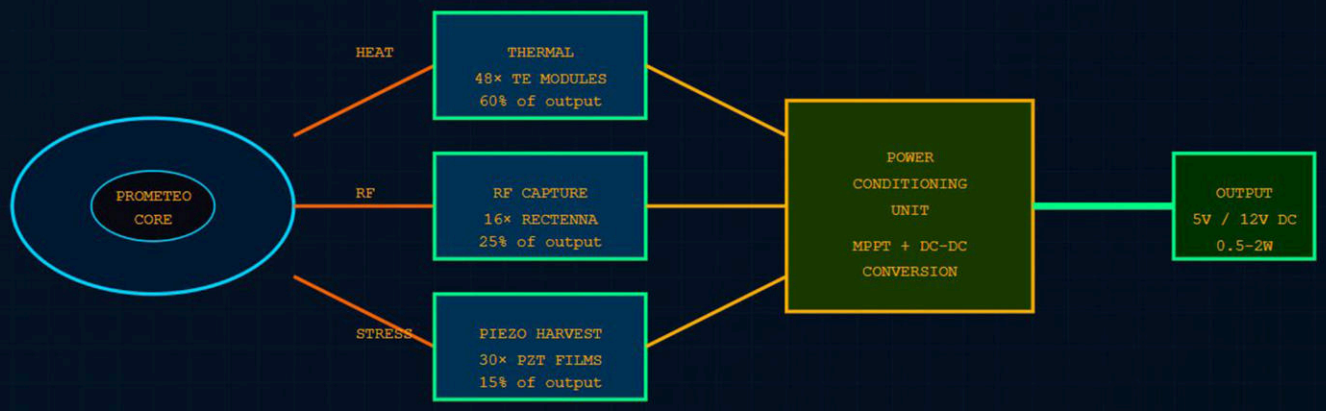
DETAIL A: THERMOELECTRIC MODULE MOUNTING — SCALE 2:1



DETAIL B: RF RECTENNA ELEMENT — SCALE 5:1



ENERGY HARVESTING SYSTEM SCHEMATIC



SUBSYSTEM	QTY	EXPECTED INPUT	EFFICIENCY	EXPECTED OUTPUT
Thermoelectric (Bi_2Te_3)	48	6-30 W thermal	5-8%	0.4-2.0 W
RF Rectenna (0.1-10 MHz)	16	0.1-0.4 W RF	40-60%	0.05-0.2 W
Piezoelectric (PZT-5H)	30	0.5-1.0 W mechanical	10-15%	0.05-0.15 W
TOTAL	—	—	—	**0.5-2.4 W**

REVISION HISTORY
REV A — 2026-03-07 — INITIAL RELEASE

DESIGNED: [REDACTED]
CHECKED: [REDACTED]
APPROVED: GHOST PROJECT LEAD

AETHERION MARK 2-V "PROMETEO"

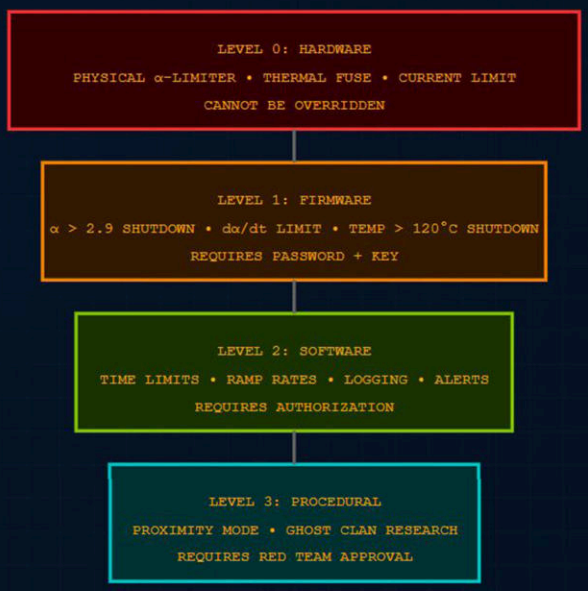
SAFETY SYSTEMS AND CONTROL ARCHITECTURE

DRAWING: RTM-UFF-AP2V-SC-004
SHEET 4 OF 4
SCALE: N/A

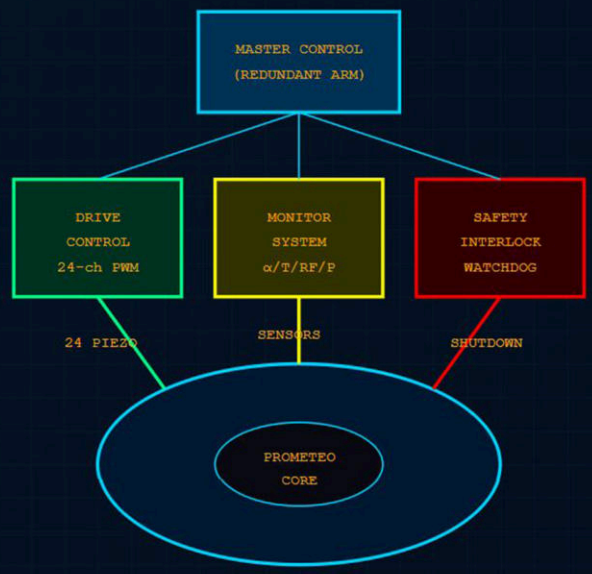
⚠ GHOST CLAN PROXIMITY WARNING

Standard operation maintains α_max = 2.5 (safe margin from Ghost Clan threshold at α ≈ 3.0)
Extended operation modes (α > 2.7) require RED TEAM authorization and remote operation protocols.
MAXIMUM ESTIMATED ENERGY RELEASE IF CONTAINMENT FAILS: ~100 kJ

SAFETY INTERLOCK HIERARCHY



CONTROL SYSTEM ARCHITECTURE



MODE	α_max	POWER	AUTHORIZATION	PERSONNEL	RISK
SAFE	2.5	100%	Standard	Normal proximity	LOW
ELEVATED	2.7	150%	Supervisor	5m exclusion	MODERATE
PROXIMITY	2.9	200%	Red Team	10m exclusion	HIGH
GHOST	3.0+	???%	DO NOT OPERATE	50m exclusion	CRITICAL

INTERLOCK	TRIGGER	ACTION	RESET
α LIMIT	α > 2.9	IMMEDIATE SHUTDOWN	Manual + review
α RUNAWAY	dα/dt > 0.1/s	IMMEDIATE SHUTDOWN	Manual + review
OVER-TEMP	T > 120°C	SHUTDOWN	Auto @ T < 80°C
COOLING FAIL	Flow < 0.5 L/min	REDUCE POWER 50%	Auto when restored
RF ANOMALY	Spectrum Δ > 3σ	SHUTDOWN	Manual review

EMERGENCY PROCEDURES:
1. IF α > 3.0: AUTOMATIC SHUTDOWN INITIATES — DO NOT APPROACH
2. IF SHUTDOWN FAILS: EVACUATE 50m — NOTIFY EMERGENCY RESPONSE
3. ESTIMATED MAXIMUM ENERGY RELEASE: ~100 kJ (~25g TNT EQUIVALENT)
4. POST-INCIDENT: DO NOT POWER ON WITHOUT RED TEAM REVIEW

REVISION HISTORY
REV A — 2026-03-07 — INITIAL RELEASE

DESIGNED: [REDACTED]
CHECKED: [REDACTED]
APPROVED: GHOST PROJECT LEAD